



A Study of Real-world Audio-Visual Corpus Design and Production: A Perspective from MISP Challenges

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Abstract

With the increasing proliferation of speech-driven applications, the challenges in their deployment environments are becoming more prominent. Accordingly, audio-visual speech processing (AVSP) proposed integrating audio and visual information to enhance performance across many speech-processing tasks. In this context, the MISP challenges were organized at ICASSP 2022, 2023, and 2024, respectively. These challenges released audio-visual corpora to support four core tasks: audio-visual wakeup, diarization, speech enhancement, and recognition. The datasets have garnered attention from the global research community, with over 110 teams downloading the corpora. This paper provides a comprehensive analysis of the MISP corpus design from various perspectives, including scenario selection, recording equipment and processes, as well as manual transcription and alignment, highlighting its strengths and limitations and offering insights and recommendations for the design of future AVSP corpora.

Index Terms: Data resource, audio-visual, MISP challenge, real-world scenarios

1. Introduction

With the emergence of many speech-enabled applications, the scenarios in which these applications are used (e.g., at home and in meetings) are becoming increasingly challenging due to several factors, particularly adverse acoustic environments (far-field audio, background noises, and reverberation) and conversational multi-speaker interactions with a significant speech overlap. State-of-the-art speech processing techniques based on the single audio modality run into performance plateaus, e.g., the CHiME-8 dinner party scenario reaches a word error rate of about 25%, which is a level of performance that falls short of the deploy ability of the application. The Cocktail Party Effect [1] demonstrates the human auditory system's ability to isolate a single source in challenging acoustic environments. This is due to the brain's capacity to integrate signals from different sensory modalities [2]. The McGurk effect [3] shows the influence of visual cues on speech perception. In addition, the role of facial and lip movements in enhancing speech perception in harsh acoustic environments was demonstrated in [4, 5].

Audio-visual speech processing (AVSP) is a recently emerging topic in the speech and language processing (SLP) and image, video, and multidimensional signal processing (IVSMP) communities, where the integration of visual information has led to significant advances in several critical speech processing tasks. Specifically, within the speech wakeup task, providing visual input has shown a remarkable reduction in

false wakeup rates, especially in challenging acoustic environments [6, 7]. Visual supplements also offer critical insight into speaker identity and spatial localization during speaker diarization, effectively mitigating overlap and localization issues [8, 9, 10, 11, 12, 13, 14, 15]. Moreover, in speech recognition, visual cues not only complement acoustic information degraded by noise but also provide discriminative cues in overlap [16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27], leading to reductions in deletion and substitution errors, respectively. The essence of those achievements lies in the skillful application of powerful audio-visual fusion methods, a prominent area of research within the multimodal signal processing field.

However, the development of AVSP methodology is still in its infancy, and we are uncertain how far we are from achieving the performance standards necessary for practical implementation in real-world applications. Therefore, it is essential to conduct a comprehensive and objective benchmarking campaign of state-of-the-art algorithms in real-world acoustic environments. This effort will facilitate a thorough analysis of recent advances, the identification of critical barriers and unresolved issues, and the exploration of potential directions for future research. Yet, there is still a notable lack of large-scale public audio-visual corpora recorded in real-world scenarios. Traditional audio-visual corpora such as GRID [28], OuluVS [29], OuluVS2 [30], and TCD-TIMIT [31] have been collected in controlled environments. However, these corpora have limitations in terms of duration, number of speakers, and vocabulary size. More recently, there has been a concerted effort within the research community to collect data from various sources in real-world settings, including public media Web sites and television broadcasts. An early example of an audio-visual in-the-wild corpus is LRW [32]. This trend of collecting larger datasets has continued with subsequent collections that include material from British television programs [33, 34], TED talks [35], and others. These datasets lack sample-level metadata for further analysis, although they may contain different speakers, sentences, languages, and visual/acoustic environments.

In this context, the multimodal information-based speech processing (MISP) challenges have been initiated to establish a comprehensive benchmarking framework tailored for AVSP in real-world home scenarios. Specifically, we organized MISP 2021 [36], 2022 [37], and 2023 [36] as the signal processing grand challenge (SPGC) of ICASSP 2022, 2023 and 2024, respectively.¹ MISP 2021, 2022, and 2023 released a set of carefully designed corpora recorded in the real-world home scenario, denoted as MISP-Home, for supporting three tasks: audio-visual wakeup, speaker diarization, speech recognition, and enhancement. They have attracted extensive participa-

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¹ICASSP SPGCs typically begin the year before the conference.

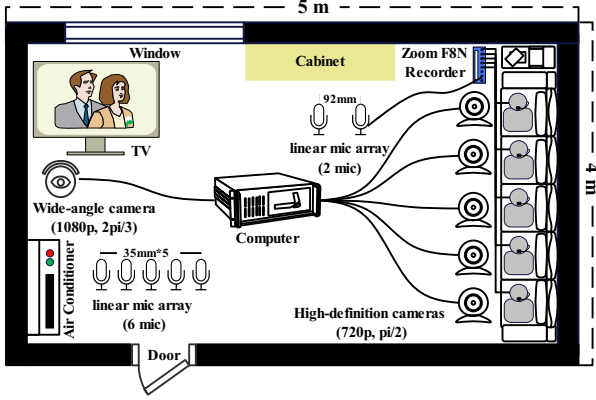


Figure 1: A schematic overview of the recording scenario and devices.

tion from the global research community, resulting in over 110 teams downloading the dataset, more than 60 teams actively submitting their results, and 10 research papers presented at ICASSP 2022 and 2023. This widespread engagement underscores the significant impact of the MISP-home dataset on AVSP. Accordingly, this paper aims to provide a comprehensive analysis of the design, recording, and processing of the MISP-Home dataset. It seeks to offer valuable insights into the future development of multimodal datasets, thereby contributing to advancements in unified multimodal intelligence [38, 39, 40, 41, 42].

This paper is structured as follows: Section 2 introduces the data recording procedure. In Section 3, we describe the devices used for data collection. Section 4 discusses the manual transcription and alignment process. Section 5 presents the statistical analysis of the MISP corpus. Finally, Section 6 offers a comprehensive analysis of the MISP corpus design and proposed improvements.

2. Recording

The MISP-Home dataset was recorded in the real-world home scenario, as illustrated in Figure 1. Specifically, between 3 and 6 participants engage in conversation while watching TV in a living room setting, with the ability to interact with a smart speaker or TV upon pronouncing specific wake-up or trigger words. Multiple microphones and cameras collected the multimodal data through the detailed recording procedure as follows:

1. **Participant Arrangement:** The participants are seated on the sofa, strategically choosing their positions to form the recording setup’s central axis.
2. **Microphone and Camera Setup:** Each participant is equipped with a high-fidelity microphone worn at the chin. A high-definition camera is positioned approximately 1.0 to 1.5 meters in front of each participant, with the height and angle adjusted to ensure their face is centered within the frame and their lips remain visible throughout the conversation. A linear microphone array consisting of two microphones is placed in front of the sofa, also at a distance of 1.0 to 1.5 meters, to capture the audio from all participants. Additionally, a linear microphone array with six microphones and a wide-angle camera are positioned around the TV, approximately 3 to 5 meters from the sofa, to capture audio-visual interactions in the broader environment.

Config ID	Time	Light	TV	Group
01	Day	off	on	1
02	Day	on	on	1
03	Day	off	off	1
04	Day	on	off	1
05	Day	off	on	2
06	Day	on	on	2
07	Day	off	off	2
08	Day	on	off	2
09	Night	on	on	1
10	Night	on	off	1
11	Night	on	on	2
12	Night	off	off	1

Table 1: Scene configuration

3. **Device Connectivity:** All cameras are connected to a central computer for synchronized data capture. The high-fidelity microphones and the two-microphone array are connected to a Zoom F8N recorder for audio recording.
4. **Configuration Setup:** The living room environment is meticulously arranged to replicate typical real-world conditions per the specified configuration.
5. **Recording Action:** After a designated action, the conversation ensues, with all devices actively recording the audio-visual interactions, capturing both verbal exchanges and non-verbal cues.
6. **End of Recording:** The recording session concludes after approximately 20 minutes, ensuring sufficient data is captured for the analysis.

Some variables in the conversation still take place in the living room; for example, the TV is turned on/off, the conversation happens during the day or night, etc. To cover the real scene comprehensively and evenly, we designed the following recording configuration by controlling variables as in Table 1. “Time” refers to the recording time; the value is day or night. “Light” refers to turning on/off the light. “TV” refers to turning on/off the TV. “Group” refers to how many groups of participants are in a conversation.

By observing the real conversations that were taking place in an actual living room, we found that the participants would be divided into several groups to discuss different topics. Grouping would result in higher overlap ratios than all participants debating the same subject. We found that average speech overlap ratios of Group = 1 and Group = 2 are 10%-30% and 50%-80%, respectively. The number of groups greater than 3 is rare when the number of participants is no more than 6.

3. Devices

According to the distance between the device and the speaker, multiple recording devices were divided into 3 categories:

1. **Far devices:** a linear microphone array (6 mic, 16 kHz, 16-bit) and a wide-angle camera (1080p, 25 fps, $2\pi/3$) placed 3-5m from the speaker. Each microphone is an omnidirectional microphone. The distance between adjacent microphones is 35mm. A profile of the microphone array is provided in Figure 2. The detailed parameters of the wide-angle camera are shown in Table 2. Additionally, we utilized checkerboard images of the intrinsic parameters to correct any distortions. The linear microphone array is fixed on the top of the wide-angle camera, parallel to the x-axis of the camera co-

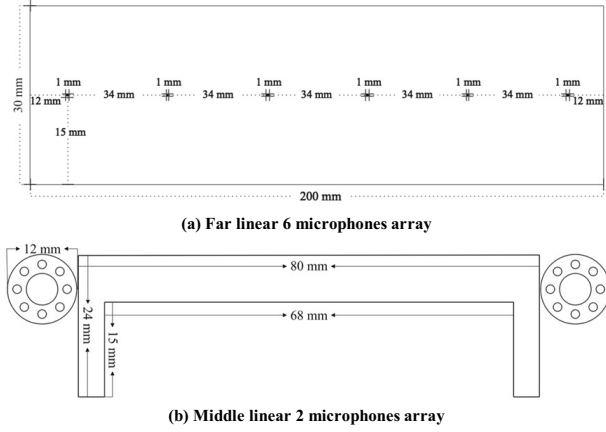


Figure 2: Profiles of (a) the far linear 6 microphones array and (b) the middle linear 2 microphones array.

Parameter	Far	Middle
Sensor type	200W CMOS Sensor	200W CMOS Sensor
Pixel size	$3\mu\text{m} \times 3\mu\text{m}$	$3.75\mu\text{m} \times 3.75\mu\text{m}$
Focus plane	1/2.7"	1/3"
Focal length	2.8mm	3mm
Aperture	$F1.8$	$F1.5$
Field of view	$D = 141^\circ$ $H = 120^\circ$ $V = 63^\circ$	$D = 116^\circ$ $H = 99^\circ$ $V = 53.4^\circ$
Resolution	1920 × 1080p	1280 × 720p
Frame rate	25fps	25fps

Table 2: Parameters of the far wide-angle and middle high-definition cameras

ordinate system, and the midpoint coincides with the origin of the camera coordinate system. All participants appear in the camera, which brings speakers position information while reducing the resolution of the lip region of interest (ROI);

- Middle devices:** a linear microphone array (2 mic, 44.1 kHz, 16-bit) and n high-definition cameras (720p, 25fps, $\pi/2$), which are placed 1-1.5m away from the speaker, where n is the number of participants within this conversation. Two microphones are omnidirectional microphones. The distance between the two microphones is 92 mm. A profile of the microphone array is provided in Figure 2. The detailed parameters of the high-definition camera are shown in Table 2. There is no relationship between the position of the linear microphone array and the high-definition camera. There is only the corresponding speaker in each camera, and the lip ROI is recorded clearly;
- Near devices:** every high-fidelity microphone (44.1 kHz, 16-bit) was stuck in the middle of the corresponding speaker’s chin, respectively. The collected audio signal is rarely interfered with by the off-target source, and the SNR is estimated to be greater than 15 dB. This provides a guarantee for high-quality manual transcription.

Figure 3 also shows real shots of various devices.

4. Manual Transcription and Alignment

The transcription process begins with skilled professional transcribers who meticulously transcribe each near-field mono



Figure 3: Real shots of various devices.

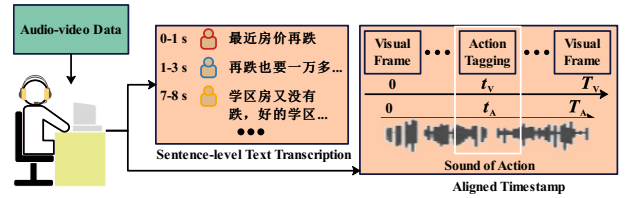


Figure 4: A schematic overview of manual transcription and alignment.

speech. As depicted in Figure 4, they listen to the audio and examine the waveform to pinpoint the precise start and end points of each sentence and transcribe the spoken content. We employ the Dacidian Dictionary as our foundational character set, requiring annotations for any words not included in this lexicon. Each transcription undergoes two rigorous rounds of quality checks, guaranteeing that timestamp errors remain within a 100-millisecond threshold and that transcription accuracy exceeds 99%.

Another critical issue is the clock synchronization among multiple devices. We address this challenge from two perspectives: synchronization devices and manual post-processing. Specifically, the sound card (ZOOM F8n) is employed to synchronize the clocks of the middle linear microphone array with those of the near high-fidelity microphones. In contrast, Vicon software, running on the industrial PC (MIC-770), is used to synchronize the clocks of all cameras.

Even with synchronization devices in place, three distinct clocks remain: the sound card clock, the clock of the far linear microphone array, and the clock of the industrial PC. These clocks are synchronized manually by identifying a specific reference point. A particular action, such as knocking a cup, is performed before recording. The visual frame, in which the cup’s wall and cover come into contact, is aligned with the corresponding waveform point that represents the impact sound.

Table 3: Overview of the MISP-Home wakeup corpus. [P: for presence of wake word, N: for absence of wake word]

Dataset	Training		Dev		Eval	Total
	P	N	P	N		
Duration (h)	5.67	112.86	0.62	2.77	2.87	124.79
Session	89	89	10	10	19	118
Room	25	25	5	5	8	38
Participant	258	258	35	35	54	347
Male	81	81	11	11	31	123
Female	177	177	24	24	23	224

Table 4: Overview of the MISP-Home recognition corpus.

Dataset	Training	Dev	Eval	Total
Duration (h)	106.09	13.32	21.83	141.24
Overlapping Duration (h)	44.57	6.47	10.70	61.74
Room	21	5	8	34
Participant	200	21	42	253
Male	79	9	10	98
Female	121	12	32	165

5. Statistics

The dataset used for audio-visual wakeup contains 124.79 hours of audio-visual data. Table 3 shows the division of the audio-visual data into a training, development, and evaluation set and indicates details regarding the number of sessions, the type of room, and the number of male/female speakers. The wake word is “Xiao T Xiao T”. The data set includes 118 sessions. The number of speakers within one conversation session ranges from 1 to 6. The total number of speakers in the data set is 347. All speakers are native Chinese speaking Mandarin without strong accents. Various conversation topics were recommended during recording. [36] shows more details.

As shown in Table 4, the MISP-Home recognition corpus for audio-visual speech recognition contains 141.24 hours of audio and video data from 253 native Chinese (98 males and 165 females) speaking Mandarin without strong accents which collected in 34 real home TV rooms, whose size range from $3.2 \times 2.56 \times 2.54$ to $5.2 \times 4.2 \times 2.8 m^3$. Each session consists of a discussion of about 20 minutes. There is no overlap in speakers and recording rooms among the data in each subset. There are multiple microphone arrays and cameras used to collect far/middle/near audio and far/middle video data, respectively. [36] shows more details.

6. Analysis

The MISP-Home dataset presents several notable advantages that contribute to its value for research in multimodal data analysis:

1. **Simultaneous Recordings from Multiple Microphone Arrays and Video Cameras:** This provides rich, synchronized data from both audio and visual sources, enabling a comprehensive analysis of multimodal interactions.
2. **Real Conversation:** The dataset captures natural, unscripted conversations where speakers talk in a relaxed, everyday manner. This makes the data highly representative of real-world communication.
3. **High Overlap Ratios (20%-40%) in Multi-Talker Conver-**

sations: The significant overlap in speech allows for more accurate testing and development of models that handle complex, interactive scenarios.

4. **Real Domestic Noise Backgrounds:** Including ambient noise from typical home environments, such as TV, air conditioning, and movement, provides a more challenging yet realistic setting for noise-resistant model training.
5. **Diverse Acoustic Conditions:** The dataset features recordings in over 30 different real room acoustics, which introduces varied challenges for speech recognition systems.
6. **Native Chinese Speakers:** The corpus consists of over 250 native Chinese speakers, providing a substantial resource for Mandarin speech recognition, particularly beneficial for developing models tailored to this language.

Despite its advantages, the MISP-Home dataset also has limitations that could hinder certain types of research:

1. **Limited Scene Variety:** The dataset primarily captures casual, unscripted conversations in a home environment, which may lack distinct, predefined topics. This lack of topic segmentation limits the applicability of the dataset for tasks such as natural language understanding (NLU), where structured dialogue is often crucial.
2. **Single Language:** The dataset only includes Mandarin, which restricts its utility for research involving multilingual or non-native Mandarin speakers. This limits the scope for researchers interested in cross-linguistic studies or multilingual model development.
3. **Equipment Limitations:** From a recording perspective, using linear microphone arrays may not fully capture the complexities of modern acoustic environments, as linear arrays are increasingly becoming outdated. Additionally, the 120-degree field of view of the cameras does not cover the entire space, which limits the ability to capture the complete spatial context of the interactions.

To address these limitations, several improvements can be made to enhance the MISP-Home dataset:

1. **Inclusion of More Diverse Scenes:** Expanding the dataset to include a variety of settings, such as meetings, would make the dataset more applicable to a broader range of research, particularly in areas like topic segmentation, summarization, key phrase extraction, and action item detection. Detailed annotations for these tasks could further enhance its value.
2. **Multilingual Support:** Providing English translations for transcriptions would make the dataset more accessible to a broader international audience. Additionally, incorporating English conversations with Chinese participants who have proven English proficiency (e.g., TEM-8 certification) would increase the dataset’s multilingual research potential.
3. **Upgrade to Advanced Recording Equipment:** The dataset can be updated with more modern equipment, including advanced microphone arrays (e.g., beamforming arrays and spherical microphones), as well as 360-degree panoramic cameras to capture a full spatial representation of the recording environment. This would significantly enhance the quality of both audio and visual data.

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